

Problem Set 4: Grasping

6.4210/2 Robotic Manipulation — Fall 2026

Due: October 11, 2026

Out: October 1 **Due:** October 11

This problem set is about contact and grasping: when a contact holds without slipping, how to estimate surface normals from a depth image, and how to use those normals — analytically and by sampling — to find antipodal grasps.

Before you start: update your Python environment. The ps4 code builds letter meshes, which needs packages that earlier ps4s did not. Pull the latest handout and check the install instructions in its `README.md` (§3.3). If you made your venv before ps4, activate it and run

```
uv pip install "manipulation[letter]" scipy rtree
```

On Colab, `setup()` installs them for you.

1 Bookwork

1.1 Contact models and wrenches

Use the notation of [Chapter 5](#). A spatial force $F_{i,W}^{B_p} = [(\tau_{i,W}^{B_p})^T, (f_{i,W}^{B_p})^T]^T$ has *torque first*, is applied at point B_p of body B , and is expressed in W . A contact frame C_i has $+z$ pointing into the object, so positive $f_{C_i,z}^{c_i}$ is a push. Surface normals used in the coding problems point *out* of the object; keep this distinction in mind. Enter the answers in the [Gradescope](#) blocks in Gradescope. The [Going further](#) prompts are for discussion and require no submission.

Problem 1 What can a contact transmit?

[3]

[Gradescope:](#)

1.1 Match each model to a description below. Use each letter once.

[1]

Models

- i. Frictionless point contact: _____
- ii. Point contact with friction: _____
- iii. Soft-finger contact: _____
- iv. Hydroelastic contact with friction: _____

Descriptions

- A. A nonnegative normal force, tangential force in a friction cone, and a bounded torsional moment about the contact normal.
- B. Contact forces over a compliant contact patch, integrated to obtain a spatial force.
- C. A nonnegative normal force only, with no moment at the contact point.
- D. A nonnegative normal force and tangential force in a friction cone, with no moment at the contact point.

1.2 For point contact with friction, select all constraints that must hold:

[1]

- ☐ $f_{C_i,z}^{c_i} \geq 0$.
☐ $\sqrt{(f_{C_i,x}^{c_i})^2 + (f_{C_i,y}^{c_i})^2} \leq \mu f_{C_i,z}^{c_i}$.
☐ The tangential force always has magnitude $\mu f_{C_i,z}^{c_i}$.
☐ $\tau_{i,C_i}^{c_i} = 0$.

1.3 True or false: because a point contact cannot apply a moment at its own location, it cannot cause angular acceleration of the object. ☐ True ☐ False

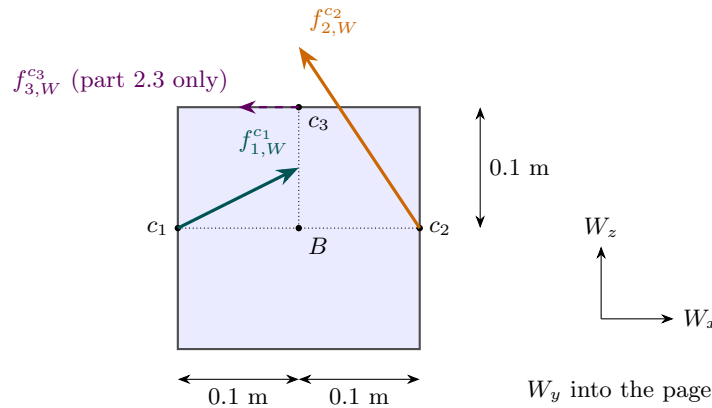
[1]

Going further: What would you want to measure before choosing a contact model for a glass marble, a rubber fingertip, or a suction cup?

Problem 2 Forces, moments, and motion

[3]

Consider a planar body moving in the world xz plane. Its center of mass is B , and the only possible rotation is about W_y . A planar spatial force keeps the components $F = [\tau_y, f_x, f_z]^T$, torque first. All data below are expressed in W ; positions are relative to B . Ignore gravity in this question. Positions and forces are given as (x, z) components; all y components are zero. Two point contacts are at $(-0.1, 0)$ and $(0.1, 0)$ meters, and apply pure forces $f_{1,W}^{c_1} = (2, 1)$ and $f_{2,W}^{c_2} = (-2, 3)$ newtons.



Gradescope:

- 2.1 Enter $[\tau_{1,W,y}^B, \tau_{2,W,y}^B, \tau_{\text{total},W,y}^B]$ in N m, and the force components $[f_{\text{total},W,x}^B, f_{\text{total},W,z}^B]$ of $F_{\text{total},W}^B$ in N. [1]

$$[\tau_{1,W,y}^B, \tau_{2,W,y}^B, \tau_{\text{total},W,y}^B] = [\text{_____}, \text{_____}, \text{_____}] \text{ N m},$$

$$[f_{\text{total},W,x}^B, f_{\text{total},W,z}^B] = [\text{_____}, \text{_____}] \text{ N}.$$

- 2.2 The body starts at rest. Does its center of mass initially accelerate, does it initially have angular acceleration, both, or neither? [1]

Answer (translation only / rotation only / both / neither): _____.

- 2.3 A third contact at $(0, 0.1)$ meters applies $f_{3,W}^{c_3} = (-1, 0)$ N. Enter the new total spatial force $F_{\text{total},W}^B = [\tau_y, f_x, f_z]^T$. [1]

$$F_{\text{total},W}^B = [\text{_____}, \text{_____}, \text{_____}]^T \text{ (N m, N, N)}.$$

Going further: How would you choose contact locations if translation were useful but rotation were undesirable? What additional information would you need to predict the motion a second later, rather than just the initial acceleration?

1.2 Friction and equilibrium

Problem 3 A box on an inclined plane

[3]

This is based on Exercise 5.1a in the book. A box of mass m rests on a plane inclined at angle γ , as in Figure 1. Assume $0 \leq \gamma < \pi/2$, and that the box does not tip. The resultant plane reaction acts at a center of pressure C somewhere along the base; its line of action passes through the center of mass in equilibrium. Write $f_{C,z}^{B_C}$ for its normal component and $f_{C,x}^{B_C}$ for its up-slope tangential component. These are sums over the contact patch.

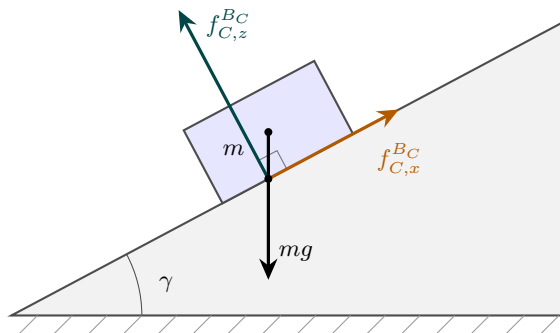


Figure 1: A box of mass m at rest on a plane inclined at γ . Gravity acts at the center of mass; the plane's reaction acts at the center of pressure, and is resolved into the normal component $f_{C,z}^{BC}$ and the tangential (friction) component $f_{C,x}^{BC}$.

Gradescope:

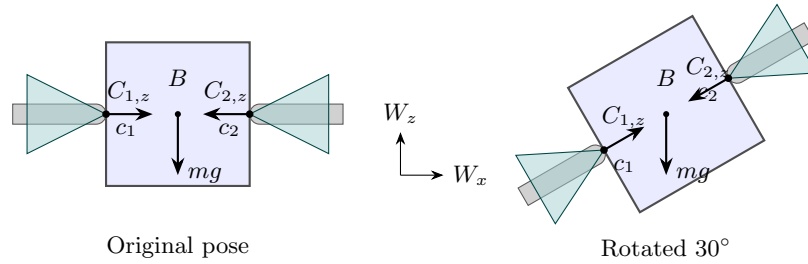
- 3.1 What is the expression for the friction force $f_{C,x}^{BC}$? [1]
- A. mg
 - B. $mg \cos \gamma$
 - C. $mg \sin \gamma$
- 3.2 What is the expression for the normal force $f_{C,z}^{BC}$? [1]
- A. mg
 - B. $mg \cos \gamma$
 - C. $mg \sin \gamma$
- 3.3 For a given angle γ , what is the minimum coefficient of friction required for the box to not slip down the plane? [1]
- A. $\mu \geq 0$
 - B. $\mu \geq \sin \gamma$
 - C. $\mu \geq \cos \gamma$
 - D. $\mu \geq \tan \gamma$
 - E. $\mu \geq \cot \gamma$

Going further: How would you decide whether a tall box will slide or tip first? What might change if the base were compliant, or if the friction coefficient varied along the ramp? Which measurements would help distinguish these effects?

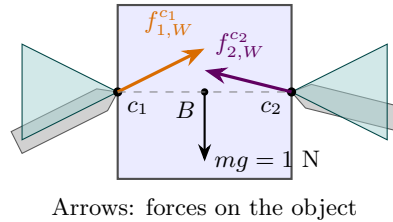
Problem 4 Holding a block between two fingers

[5]

A square of side $2a = 0.2$ m has its center of mass at its center B and weight $mg = 1$ N. Work in the xz plane, with W_z upward; positions and forces are given as (x, z) components. Ideal point fingers touch the midpoints of the left and right faces, with $\mu = 1/2$. The green friction cones are drawn *outward* from the object, along its outward surface normals. They represent admissible (consistent with the friction cone) forces on the fingers; the corresponding forces on the object point in the opposite directions. Arrows labeled $C_{i,z}$ show the inward contact-frame normals used in the equations. Gray shapes depict fingers; treat their tips as point contacts.

**Gradescope:**

- 4.1 What is the full opening angle of each friction cone, in degrees to two decimal places? _____°. [1]
- 4.2 In the original pose, each finger supplies a *normal component* of exactly 1 N; its tangential component is whatever equilibrium requires. Can the grasp balance gravity? Enter yes/no and the upward tangential force at each finger. Is the solution strictly inside the friction cones or on their boundaries? [1]
- Balance gravity (yes/no): _____.
- Upward tangential force: left _____ N; right _____ N.
- Cone location (interior/boundary): _____.
- 4.3 Now rotate the square and its two midpoint contacts counterclockwise by 30° in the drawing. The contact normals rotate with the faces. Each normal component is still exactly 1 N. Can these contacts balance gravity? (yes/no): _____. [1]
- 4.4 Return the block to the original pose. The left finger's total force is fixed at $f_{1,W}^{c1} = (1, 0.5)$ N, while the right finger's total force is $f_{2,W}^{c2} = (-1, 0.25)$ N. These are asymmetric force directions, *not* rotated contact normals. Are both forces admissible? Enter the total spatial force $F_{\text{total},W}^B = [\tau_y, f_x, f_z]^T$ about B , including gravity, and say whether static equilibrium holds. [1]



Both forces admissible (yes/no): _____.

$$F_{\text{total},W}^B = [\text{_____, _____, _____}]^T \quad (\text{N m, N, N}).$$

Static equilibrium (yes/no): _____.

- 4.5 With frictionless contacts at these same two face midpoints, can any choice of compressive normal forces hold the block against gravity? (yes/no): _____. [1]

Going further: Which would you change first to make this grasp reliable: squeeze force, contact locations, fingertip material, or sensing? How would your choice change for a fragile object? Would changing the finger approach directions change the contact normals if the contacts stay on the same flat faces?

Problem 5 Force closure is more than holding against gravity

[3]

This problem continues Problem 4: the same square block, held at the midpoints of its left and right faces by two point fingers with $\mu = 1/2$. Work in the same vertical xz plane, with translation in x and z and rotation about W_y . A planar spatial force has components $F = [\tau_y, f_x, f_z]^T$. Let $\mathcal{K}_{\text{total},W}^B$ be the Minkowski sum of the contact wrench cones, all shifted to B and expressed in W . Force closure means

that this cone is the entire planar wrench space $\mathbb{R}^3$. For this idealized definition, normal forces can grow without bound and each finger can independently choose an admissible force. A real force limit instead bounds the disturbances that can be resisted.

Gradescope:

5.1 Does the original-pose grasp from Problem 4 have *planar* force closure when the fixed 1 N normal-force restriction is removed? (yes/no): _____ [1]

5.2 In the plane, for each contact model below, can force closure be achieved with *some suitable object and contact placement* using two, three, or four contacts? Enter a row of three yes/no answers for each model. Additional contacts may apply zero force; this is an existence question, not a guarantee for arbitrary placements. [1]

Contact model	Two contacts	Three contacts	Four contacts
With friction ($\mu > 0$)	_____	_____	_____
Frictionless	_____	_____	_____

5.3 Four frictionless fingers touch the four face midpoints of a square. Does this arrangement have planar force closure? (yes/no): _____ [1]

Going further: For what tasks would you accept a grasp without force closure? How might a table, gravity, or a second hand help? What does the ideal force-closure test leave out about how a real hand generates and controls contact forces?

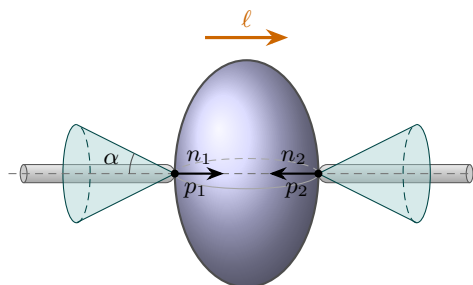
Problem 6 Antipodal grasps and force closure

[3]

Two fingers touch a rigid object in 3D at p_1 and p_2 . The inward surface normals are n_1 and n_2 , and the friction angle is $\alpha = \arctan \mu$. Let ℓ be the unit vector from p_1 to p_2 . The grasp is *antipodal* when the line connecting the contacts lies inside both friction cones:

$$\angle(\ell, n_1) \leq \alpha \quad \text{and} \quad \angle(-\ell, n_2) \leq \alpha.$$

This lets the fingers apply equal and opposite squeezing forces along the line without slipping at the contacts. For this question, assume the line lies strictly inside both cones. As in the earlier figures, the green cones point outward; the normals n_i point into the object.



A spatial two-finger grasp

Gradescope:

6.1 With hard point contacts and friction, which wrench can the grasp never resist, no matter how hard the fingers squeeze or how large μ is? [1]

- A. A force along ℓ
- B. A force perpendicular to ℓ
- C. A torque about ℓ
- D. A torque about an axis perpendicular to ℓ

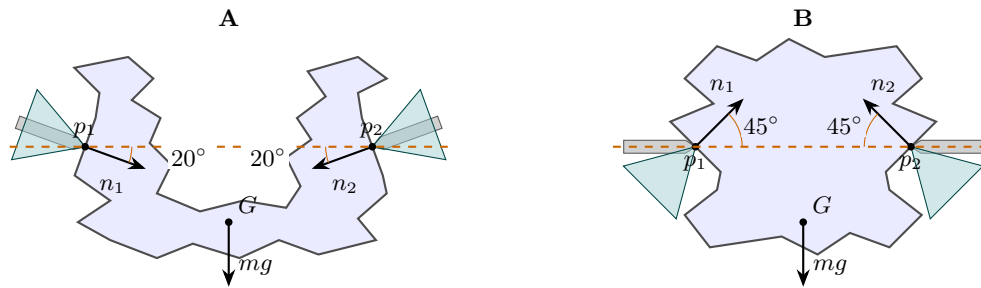
6.2 What is the minimum number of hard point contacts with friction needed for force closure in 3D? [1]

6.3 Suppose each contact is a soft finger and can also apply a moment about its normal, with $|\tau_i| \leq \mu_t f_{n,i}$ and $\mu_t > 0$. Does this antipodal grasp have force closure? (yes/no): [1]

Problem 7 Antipodal grasps on non-convex objects [6]

Consider the two rigid objects below in the vertical xz plane, with $+x$ to the right and $+z$ upward. Each has weight $mg > 0$ and center of mass G directly below the midpoint between the contacts. The contacts are hard point contacts with $\mu = \tan 30^\circ$. Each finger can independently apply any force in its friction cone, with no upper bound on its normal force.

Each contact lies on a flat facet, not at a corner. The black arrows are inward normals, the green cones point outward, and the dashed orange line joins the contacts. All labeled angles are exact.



Gradescope:

7.1 Is the contact pair in A antipodal? (yes/no): [3]

- Can these contacts hold the object in static equilibrium against gravity? (yes/no):
- Do these contacts have planar force closure? (yes/no):

7.2 Is the contact pair in B antipodal? (yes/no): [3]

- Can these contacts hold the object in static equilibrium against gravity? (yes/no):
- Do these contacts have planar force closure? (yes/no):

Going further: In A, part of the line between the contacts lies outside the object. Does that matter for whether the pair is antipodal? In B, what do the local surface normals tell you that the finger directions do not? Suppose the estimated normals are off by a few degrees. Which grasps would still be antipodal? How would curved surfaces affect your answer if the fingers landed slightly away from the planned contact points?

Problem 8 What should a grasp-quality score measure? [4]

A grasp planner can produce thousands of candidate grasps, so we want a single number that ranks them. This problem builds one standard score in four steps and then asks what it does and does not measure. Stay in the plane of Problem 5: the vertical xz plane, with translation in x and z and rotation about W_y .

1. From cones to finitely many forces. A friction cone holds infinitely many forces. Replace it with a few *generator* forces along its edges, each scaled to a unit normal component. In the plane, a cone has exactly two edges, so its two edge forces describe it exactly.

2. From forces to wrenches. Shift each generator to B , as in Problem 5. Generator j at contact i becomes a wrench $F_{i,j,W}^B = [\tau_y, f_x, f_z]^T$. Nonnegative combinations of these wrenches give exactly the cone $\mathcal{K}_{\text{total},W}^B$.

3. From a cone to a shape with a size. A cone has no size: if it contains a wrench, it contains every positive multiple of that wrench. Instead, take the *wrench hull* $\mathcal{H} = \text{conv}\{F_{i,j,W}^B\}$, the smallest

convex set containing every generator wrench. Roughly, $\mathcal{H}$ holds the wrenches that the fingers can apply together when they share one unit of total normal force. A score then measures how far $\mathcal{H}$ extends from a chosen point.

4. Units. A distance in wrench space mixes Nm with N. Choose a fixed characteristic length ℓ , such as the object's size, and measure in the coordinates $[\tau_y/\ell, f_x, f_z]$, all in newtons.

The parts ask when $\mathcal{H}$ certifies force closure, which measurement of $\mathcal{H}$ is a sensible score with and without gravity, and what is cheap to compute.

Gradescope:

- 8.1 Select all correct statements. [1]
- ☐ The origin strictly inside the full-dimensional hull certifies force closure for the cone generated by these wrenches.
 - ☐ The origin anywhere in the hull, including its boundary, is enough.
 - ☐ A nonempty hull certifies force closure.
 - ☐ Balancing gravity at the specified pose certifies force closure.
- 8.2 Which score measures a worst-direction disturbance margin about zero? [1]
- A. The volume of the hull.
 - B. The radius of the largest ball *centered at the origin* contained in the hull.
 - C. The radius of the largest ball, centered anywhere, contained in the hull.
- 8.3 For a gravity-loaded grasp, where should the ball instead be centered to measure *additional* disturbance margin? [1]
- A. At the origin.
 - B. At $F_{g,W}^B$.
 - C. At $-F_{g,W}^B$.
 - D. At the centroid of the hull.
- 8.4 Select all correct statements about practical computation. [1]
- ☐ Normal/chord alignment is a cheap geometric filter for a parallel-jaw pinch, but does not test reachability or collision.
 - ☐ Every good dexterous-hand grasp must have two opposed, aligned normals.
 - ☐ Given fixed contacts and linearized cones, force balance with bounds can be tested by a linear feasibility program.
 - ☐ Constructing a high-dimensional hull is always as cheap as taking two dot products.

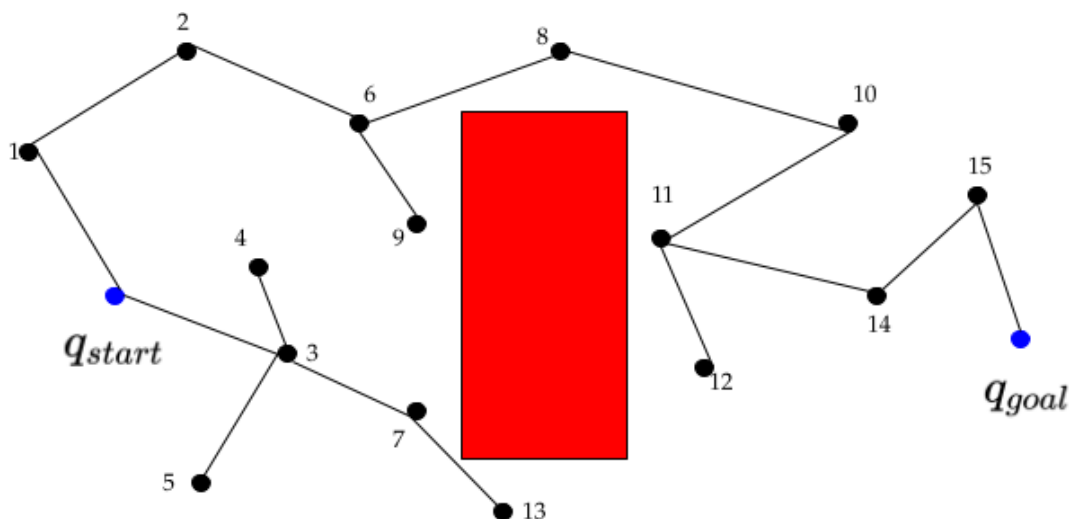
Going further: Design a sequence of inexpensive and expensive tests for thousands of candidate grasps. Which failures should be rejected early? How would you compare a large wrench margin with a short, collision-free approach when friction and center of mass are uncertain? How would you choose the torque scale ℓ ?

1.3 Motion planning

Problem 9 Shortcutting

For this problem, work through Exercise 6.3 in the textbook on improving RRT path quality. Path quality means path length, measured as Euclidean distance, and the extend operator is a straight line. [3]

- 9.1 Below is a tree grown by RRT, in which the path found from q_{start} to q_{goal} is rather jerky. Shortcutting picks two non-consecutive nodes on that path, tries to connect them with a straight line, and keeps the result if it is collision-free and shorter. Select each pair of nodes for which the shortcutting algorithm would result in a shorter path to the goal. [1]

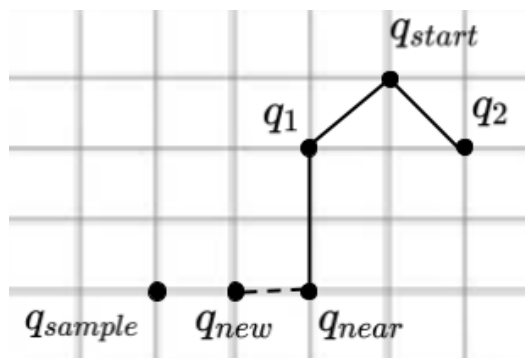


The tree, with the obstacle in red and q_{start} and q_{goal} in blue. Distances are Euclidean in the plane of the figure. (The graph was drawn by hand as an illustrative example.)

- ☐ (2, 8)
- ☐ ($q_{start}, 13$)
- ☐ (1, 6)
- ☐ (4, 9)
- ☐ (8, 15)
- ☐ (12, q_{goal})

9.2 Now zoom in on one iteration of RRT, shown below: q_{sample} is the randomly generated configuration, q_{near} the closest node in the existing tree, and q_{new} the extension from q_{near} toward q_{sample} . When the standard RRT algorithm adds q_{new} to the tree, what node is its parent?

[1]



One extension step. The solid edges are already in the tree; the dashed edge is the extension. Grid cells are unit squares.

- A. q_1
- B. q_2
- C. q_{near}
- D. q_{start}
- E. q_{sample}

9.3 If we wanted our tree to encode the shortest path from the starting node, q_{start} , to each node in the tree, what node should be the parent node of q_{new} ? Consider only existing nodes within 3 grid units of q_{new} ; keep the existing tree edges fixed and assume candidate connections are collision-free.

[1]

- A. q_1
- B. q_2
- C. q_{near}

- D. q_{start}
- E. q_{sample}

2 Computational questions

2.1 Grasping

Problem 10 Normal estimation from a depth image

[6]

Work in `normal_estimation.py`. We use neighboring pixels of a depth image to find nearby points, avoiding a separate nearest-neighbor search. The scene contains a mustard bottle, and the given mask identifies its pixels. The helper `load_camera_data()` returns the depth image, mask, camera intrinsics, and camera pose ${}^W X^C$. Use the calls in the `__main__` block at the end of the file to run the steps below.

- 10.1 Run `show_input()` to see the bottle's depth image, with a colorbar in meters and black outside the mask. Then run `illustrate_window()` to see the estimator's pixel windows outlined in red, zoomed in on the bottle. Close each plotting window to continue. We will estimate a surface normal from each patch of points.

Study `project_depth_to_pC`, which takes rows (v, u, Z) and the camera intrinsics. Then read `depth_points_in_camera_frame`, which takes a depth image and the intrinsics and returns an $(H, W, 3)$ array of camera-frame points. Image arrays are indexed by row and column (v, u) ; u increases rightward and v downward. A depth Z maps to

$${}^C p^P = \begin{bmatrix} (u - c_x)Z/f_x \\ (v - c_y)Z/f_y \\ Z \end{bmatrix}.$$

These points are expressed in the camera frame C , not in world coordinates.

- 10.2 For a window containing points $p_i = {}^C p^{P_i}$, define

[1]

$$\bar{p} = \frac{1}{n} \sum_i p_i, \quad W = \sum_i (p_i - \bar{p})(p_i - \bar{p})^T.$$

W is the 3×3 scatter matrix of the window's points: each eigenvalue measures how spread out the points are along its eigenvector. Suppose the points lie on a plane and are spread out in two directions within it, not bunched along a line or at a single point (a *nondegenerate* patch: only then does it pick out one plane).

Gradescope: For such a patch, which eigenvalue of W has the eigenvector that estimates the normal? Select one.

- A. Smallest eigenvalue.
- B. Middle eigenvalue.
- C. Largest eigenvalue.

Does a matrix whose columns are three unit eigenvectors always represent a proper rotation?

Answer (yes/no): _____.

10.3 Implement `estimate_normal_by_nearest_pixels`. There is nothing to submit for this part: part 10.4 grades it. Use `normal_patches(pC, mask, uv_step)`, passing through the supplied `uv_step`. It uses `scanning_windows` to visit masked centers on the mask's bounding-box grid in row-major order. With the defaults `UV_STEP=10` and `WINDOW_HALF_WIDTH=3`, the windows are 7×7 before clipping to the image bounds. Each patch contains only masked points with finite coordinates and positive depth. Skip patches with fewer than three points or with second-smallest eigenvalue at most 10^{-12} (in m^2). This rejects patches that do not determine a plane normal. Use `np.linalg.eigh` and orient the normal toward the camera: in C , choose its sign so that $n \cdot \bar{p} \leq 0$, with both n and $\bar{p}$ expressed in C . Construct a right-handed frame N with this normal as $+z$. Put its origin at the patch centroid. Return the list of world poses ${}^W X^N = {}^W X^C {}^C X^N$ in iterator order. Run `test1()` to display the normal frames, camera frame, and camera frustum in Meshcat. It also prints the number of normal frames. We grade each frame's origin and $+z$ axis; rotation about the normal, and hence the choice of tangent axes, is arbitrary.

10.4 Run `test2()`. This evaluates your estimator on a fixed synthetic 3×3 patch without loading the scene or starting Meshcat, so the graded answer does not depend on a renderer. [5]

Gradescope: Enter the printed synthetic frame's translation and $+z$ axis, both expressed in W . Use x, y, z component order and four decimal places.

$${}^W p^N = [\text{_____}, \text{_____}, \text{_____}]^T \text{ m.}$$

The $+z$ axis of N , expressed in W , is

$$[\text{_____}, \text{_____}, \text{_____}]^T.$$

Problem 11 Sampling-based antipodal grasping

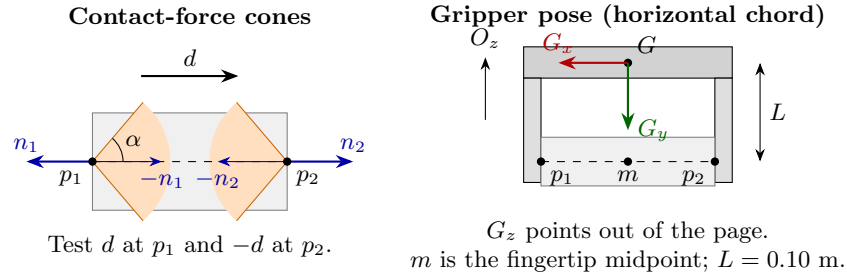
[8]

Work in `sampling_antipodal_grasps.py`. We are given a mesh in object frame O and its true world pose ${}^W X^O$. The robot should pick up the first letter, turn it to match the orientation of the second, and put it down. Start with the provided pair of B's; try your own initials after that works. The letter meshes are generated when the code runs, which needs the letter-mesh packages: in a local venv run `uv pip install "manipulation[letter]"` `scipy` `rtree` (Colab's `setup()` installs them).

11.1 Run `test1()` to inspect the scene. Read the station and controller code. Then study how the tracking diagram is assembled in `build_system_tracking_keyframes`. The controller uses the pseudo-inverse Jacobian as in PS2. It does not enforce joint limits or plan collision-free paths.

11.2 Implement `sample_colinear_points`. First sample the mesh surface using `trimesh.sample.sample_surface`. Obtain outward normals from the sampled face indices. From each point, cast a ray into the object along the negative normal. Use a small inward offset to avoid hitting the starting face. To cast the ray, call `mesh.ray.intersects_location(ray_origins, ray_directions)`, which returns every point where each ray crosses the mesh, the index of the ray it came from, and the index of the face it hit ([API documentation](#)). The arguments are arrays of shape $(n, 3)$, so a single ray is one row. These intersections are unordered: keep the nearest positive-distance hit and its face normal. Return tuples (p_1, p_2, n_1, n_2) , all in O . Pass the supplied random generator as the sampling seed. Read the function docstring for tolerances.

- 11.3 Implement `compute_grasp_from_points` and `get_filtered_grasps`. The first turns one candidate into a gripper pose in O ; the second filters the candidates and calls the first on each survivor. Parts 11.4 and 11.5 grade this part and the last one. Each candidate is (p_1, p_2, n_1, n_2) in object frame O , with unit *outward* normals n_1, n_2 . Here μ is the dimensionless coefficient of static friction assumed by our grasp filter. Use the supplied `mu` argument; its default is `LETTER_MU_STATIC = 1.17`. The inward contact-force cone has half-angle $\alpha = \tan^{-1} \mu$, measured from $-n_i$.



compute_grasp_from_points: construct the pose in O . The gripper's x axis is its closing axis; $+y$ points from its body toward the fingertips. Let width $w = \|p_1 - p_2\|$ and midpoint $m = (p_1 + p_2)/2$. Let k be the upward unit vector along the object frame's positive z axis, expressed in O . Return `None` if $w < 10^{-9}$ m. Also return `None` if the chord is nearly vertical: its unit direction has a horizontal component with norm less than 10^{-6} . Choose $+G_x$ from p_2 toward p_1 , and choose $+G_y$ as close to downward as orthogonality permits. Complete a right-handed frame. Use these axes as the *columns* of ${}^O R^G$ and return a `RigidTransform` with an appropriate translation. Letting $L = \text{FINGER_LENGTH} = 0.10$ m, the pinch point has coordinates $[0, L, 0]^T$ in G . The letters lie flat, so $O_z = W_z$ in this scene.

get_filtered_grasps: filter candidates. The central test is that the chord lies in both friction cones. For $d = (p_2 - p_1)/w$, check

$$(-n_1)^T d \geq \cos \alpha, \quad (-n_2)^T (-d) = n_2^T d \geq \cos \alpha.$$

Also require

$$\min_pt_dist \leq w \leq \max_pt_dist, \quad |n_1^T k| \leq \text{z_axis_thresh}.$$

The last test rejects contacts whose first normal is too close to vertical; it is applied to n_1 only. Reject coincident points before dividing by w , and skip candidates for which `compute_grasp_from_points` returns `None`.

get_filtered_grasps: test for collisions. Transform each surviving pose by ${}^W X^G = {}^W X^O {}^O X^G$, then call the given `check_collision_free`. It checks that all eight corners of a gripper-body box have $z_W > 0$; it does not test finger/mesh, arm, or approach-path clearance. Return a list of accepted world poses in input order (an empty list if none survive).

Run `test2()` to visualize a handful of surviving poses. Would these all work effectively to pick up the object?

- 11.4 Implement `grasp_moment_arm`. Given the gripper body pose X_{WG} and the object's center of mass p_{Wcom} , expressed in world coordinates, return the horizontal distance between the pinch point and the center of mass, in meters. Measure this distance in the world x - y plane. Recall that the pinch point is at $[0, L, 0]^T$ in G ; it is not the gripper body origin.

Why this matters: with two ideal point contacts, the grasp cannot resist torque about the line joining the contacts, which is the G_x axis. We therefore prefer grasps with the pinch point close to the object's center of mass, to reduce the torque caused by gravity.

The given `sample_grasp` selects the candidate with the smallest value. This minimizes the magnitude of the gravitational torque about the pinch point; it is a heuristic for grasp selection, not a guarantee of stability.

Gradescope: Run `test3()`, which uses fixed geometry. Enter the reported lengths to four decimal places, and report whether the grasp passes the table-clearance test.

[2]

Grasp width: _____ m.
 Moment arm: _____ m.
 Table clearance (yes/no): _____.

- 11.5 Implement `plan`. Return the keyframes to approach with open fingers, close while stationary, lift, turn, lower, open while stationary, retreat, and return to the initial pose. Each keyframe is a pair `(X_WG, opening)`: a gripper pose, and the finger opening to command there, which is always one of the constants `OPENED` or `CLOSED` defined in the script. Get the poses from the given `compute_prepick_pose` (the approach and retreat) and `lift` (up by `LIFT_HEIGHT`, and back down by passing `-LIFT_HEIGHT`), and turn by `TURN_ANGLE`; the turn is a positive 30° rotation about G_y , which points downward for an overhead grasp, hence a clockwise turn viewed from above. Compose the turn about the *pinch point* so it does not translate during this in-place rotation. Run `test4()`, which should report zero pinch displacement and a -30° world yaw change, then `test5()`.

[6]

Set `INITIALS` to your own two initials and run `test5()`. Inspect the motion and try another grasp if the object slips or the pick-and-place fails.

Gradescope: Submit a video showing the robot picking up your first initial, turning it to match the orientation of the second, placing it back on the table, and retreating. Both initials should be visible, and the video should show the complete motion.

Play the simulation back in Meshcat in your browser and take a screen recording. Upload it to Gradescope as an **mp4 file**, much smaller than 500 MB. Gradescope should indicate “Submitted”; the points will be assigned when the staff grades the video.

2.2 Additional problems for 6.4212

Problem 12 Analytic antipodal grasping 6.4212 only

[5]

Work in `analytic_antipodal_grasps.py`. Run `test1()` and read `symbolic_derivatives_demo`, which demonstrates Drake’s symbolic Jacobian and Hessian. The supplied `shape(t)` describes a smooth gear boundary $p(t) = {}^O p^{P(t)}$ in a planar object frame.

Here we seek distinct points with a chord normal to the boundary at both ends. This is the normal-aligned special case of an antipodal grasp; with friction, a chord can lie inside both friction cones without being exactly normal. On a nonconvex boundary, stationarity alone does not even guarantee that the chord points into the object at both ends. Define

$$E(t_1, t_2) = \frac{1}{2} \kappa \|p(t_1) - p(t_2)\|^2, \quad \kappa = 2.$$

- 12.1 **Gradescope:** For a normal-aligned antipodal pair, why is $\nabla E = 0$? Select one.

[1]

- A. The chord is tangent at both ends.
- B. The chord is orthogonal to both tangents.
- C. The points coincide.

- 12.2 **Gradescope:** Select all reasons that zero gradient alone can fail to certify a usable grasp.

[1]

- ☐ The contact points coincide.
- ☐ A boundary tangent vanishes.
- ☐ The chord points outward at one endpoint.
- ☐ The gripper collides during its approach.

- 12.3 Implement `find_antipodal_pts` using symbolic derivatives and `MathematicalProgram`. This is a feasibility problem, not minimization:

[1]

$$\text{find } (t_1, t_2) \text{ s.t. } \nabla E = 0, \quad 0 \leq t_i \leq 2\pi - \varepsilon, \quad t_1 - t_2 \geq \varepsilon, \quad \varepsilon = 10^{-3}.$$

Use at most `max_attempts` random initial guesses from the supplied `rng`. A successful solver status is not enough: verify the residual, parameter bounds, nonzero tangents, distinct points, and inward-facing chord at both contacts using `is_antipodal_pair`. This given helper assumes a simple closed

boundary; **orientation** is +1 for counterclockwise parameterization and -1 for clockwise (the gear uses -1). Return the accepted parameters and the sorted eigenvalues of the Hessian, using `np.linalg.eigvalsh`. Raise `RuntimeError` if no pair is found. Run `test2()` to inspect a gear grasp.

Gradescope: Is this a convex optimization problem? (yes/no): _____.

12.4 Run `test3()` to analyze fixed stationary pairs on an ellipse $p(t) = (2 \cos t, \sin t)$, using the same $\kappa = 2$. It prints the Hessian's eigenvalues at each pair. Use them to classify the pair. [1]

12.4.1 **Gradescope:** Pair $(t_1, t_2) = (0, \pi)$ is a:
 A. minimum B. maximum C. saddle D. inconclusive

12.4.2 **Gradescope:** Pair $(t_1, t_2) = (\pi/2, 3\pi/2)$ is a:
 A. minimum B. maximum C. saddle D. inconclusive

12.5 **Gradescope:** At a stationary pair, select all valid conclusions. [1]

- ☐ A positive-definite Hessian gives a strict local minimum of E .
- ☐ A local maximum of E means the boundary is concave at both contacts.
- ☐ A zero Hessian eigenvalue makes this second-derivative test inconclusive.
- ☐ Minimizing this energy alone certifies a stable grasp by the real robot.

Run `test4()` to look for examples of different Hessian signatures on the gear. Discuss what this energy model leaves out about friction, finite fingertips, and the motions permitted by a real gripper.

3 What you should understand about the code you just produced

You should be able to answer these questions without running the code, using a short explanation, sketch, or calculation. Written tests may ask you to apply these ideas to a different geometry or situation.

- How does a depth discontinuity affect PCA? What changes when a window contains a narrow strip of points rather than a surface patch?
- Two implementations return opposite normal vectors for the same patch. Can both be valid PCA results? How do we choose the sign? Can two valid normal frames still have different tangent axes?
- The camera is translated without changing its orientation. How does transforming a patch centroid from camera coordinates into world coordinates differ from transforming its normal? What goes wrong if you transform both as points?
- What is the role of comparisons with `min_pt_dist` and `max_pt_dist` in the grasp filtering?
- Which of our grasp filters express physical constraints, and which are heuristics? What failures can survive all of them?
- Explain the signs in $(-n_1)^T d \geq \cos \alpha$. Write the corresponding test at the second contact. Why must the normals and d be unit vectors?
- What if we positioned the gripper's body origin at the midpoint between the two desired contact points? Would the fingertips reach those points? Use a sketch to explain where the body origin should be instead.
- What if we changed the gripper's orientation while keeping its body origin fixed? Would the midpoint between the fingertips stay fixed? Sketch what happens, and explain how the body's position must change to keep that midpoint fixed.
- A grasp passes our `check_collision_free` function, but the robot hits the object or table during execution. How is that possible? Describe what the function actually checks and two additional checks needed for the complete motion.
- Why can a small gravity moment arm be useful without certifying force closure?
- A grasp can support the object against gravity in its initial pose but does not have force closure. What could go wrong when the robot rotates the object? What could go wrong if it follows the same path faster? What must a planner check beyond collisions?
- **6.4212:** The solver reports success and $\nabla E = 0$. Give two reasons the returned pair might still be invalid. If the pair is valid, does the Hessian's classification tell you whether the real grasp is stable?